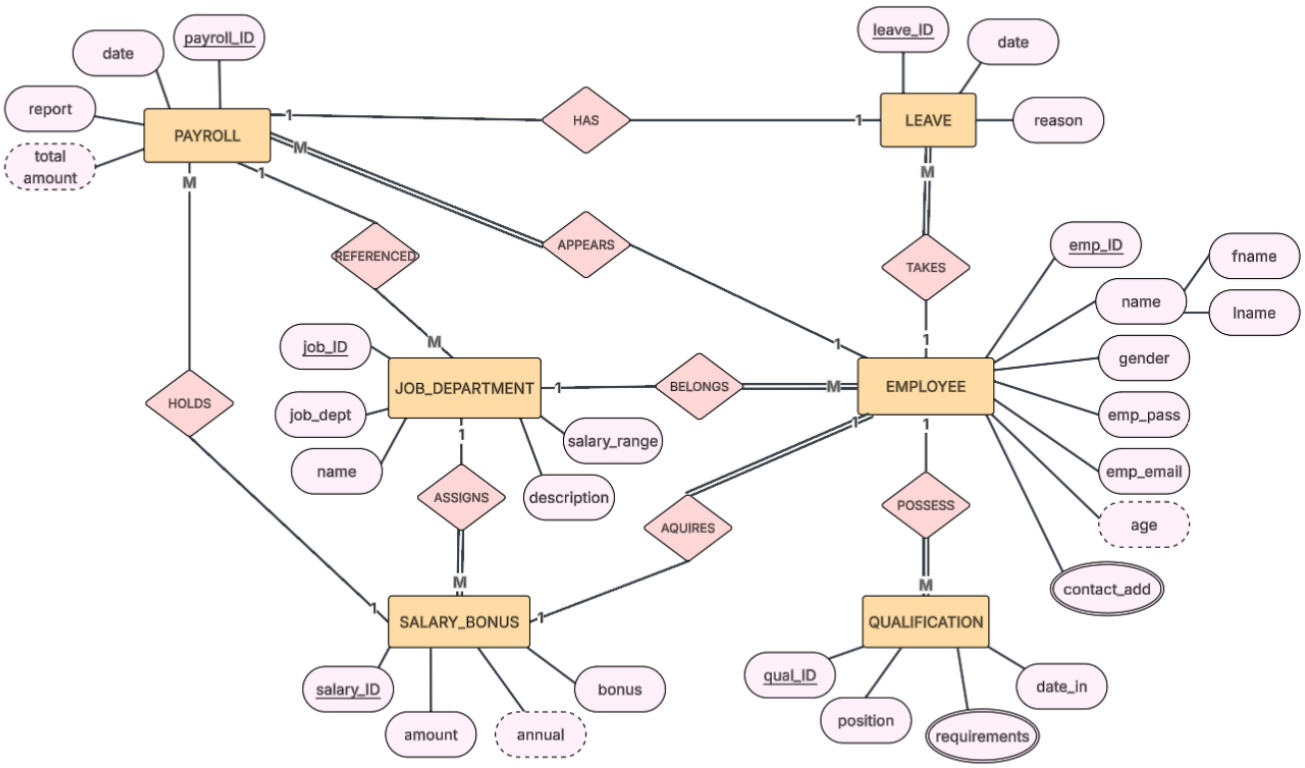
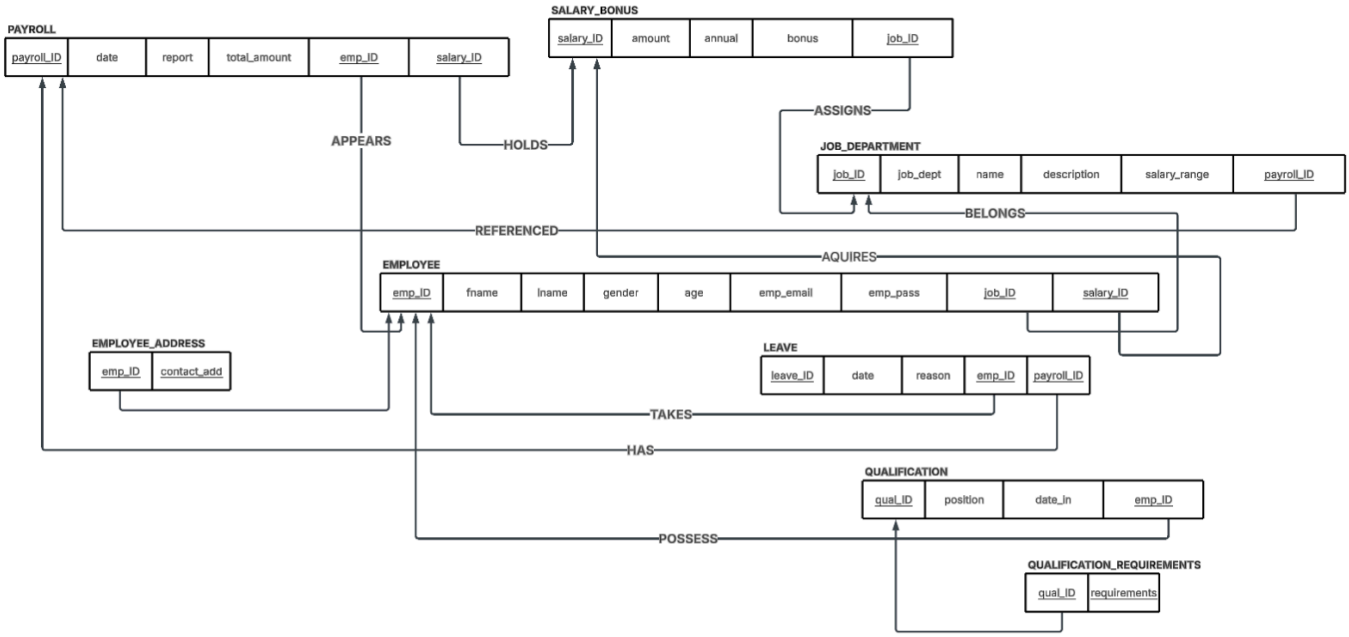
Employee Management System

01 ERD Design & Mapping

Task 1:



Task 2:



Mapping:
PAYROLL (PK_payroll_ID, date, report, total_amount, FK_emp_ID, FK_salary_ID)
SALARY_BONUS (PK_salary_ID, amount, annual, bonus, FK_job_ID)
JOB_DEPARTMENT (PK_job_ID, job_dept,name,description, salary_range, FK_payroll_ID)
EMPLOYEE (PK_emp_ID, fname,lname, gender, age, emp_email, emp_pass, FK_job_ID, FK_salary_ID)
LEAVE (PK_leave_ID, date, reason, FK_emp_ID, FK_payroll_ID)
QUALIFICATION (PK_qual_ID, position,date_in,FK_emp_ID)
EMPLOYEE_ADDRESS (PK_FK_emp_ID, PK_contact_add)
QUALIFICATION_REQUIREMENTS (PK_FK_qual_ID, PK_requirements)

Foreign Keys:
PAYROLL.emp_ID references EMPLOYEE(emp_ID)
PAYROLL.salary_ID references SALARY_BONUS(salary_ID)
SALARY_BONUS.job_ID references JOB_DEPARTMENTS(job_ID)
JOB_DEPARTMENTS.payroll_ID references PAYROLL.(payroll_ID)
EMPLOYEE.job_ID references JOB_DEPARTMENTS(job_ID)
EMPLOYEE.salary_ID references SALARY_BONUS(salary_ID)
LEAVE.emp_ID references EMPLOYEE(emp_ID)
LEAVE.payroll_ID references PAYROLL.(payroll_ID)
QUALIFICATION.emp_ID references EMPLOYEE(emp_ID)
EMPLOYEE_ADDRESS.emp_ID references EMPLOYEE(emp_ID)
QUALIFICATION_REQUIREMENTS .qual_ID references QUALIFICATION(qual_ID)

Task 3:

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-- TABLE 1: JOB_DEPARTMENT

CREATE SEQUENCE job_dept_seq START WITH 1 INCREMENT BY 1;

CREATE TABLE JOB_DEPARTMENT (
   job_ID          NUMBER DEFAULT job_dept_seq.NEXTVAL PRIMARY KEY,
   job_dept        VARCHAR2(50)        NOT NULL,
   name            VARCHAR2(100)       NOT NULL,
   description     VARCHAR2(255),
   salary_range    VARCHAR2(50),
   payroll_ID      NUMBER
);

-- TABLE 2: SALARY_BONUS

CREATE SEQUENCE salary_bonus_seq START WITH 1 INCREMENT BY 1;

CREATE TABLE SALARY_BONUS (
   salary_ID       NUMBER DEFAULT salary_bonus_seq.NEXTVAL PRIMARY KEY,
   amount          NUMBER(10,2)        NOT NULL,
   annual          NUMBER(10,2),
   bonus           NUMBER(10,2),
   job_ID          NUMBER              NOT NULL,
   CONSTRAINT chk_amount_positive CHECK (amount > 0),
   CONSTRAINT fk_sb_job FOREIGN KEY (job_ID)
      REFERENCES JOB_DEPARTMENT(job_ID)
);


-- TABLE 3: EMPLOYEE


CREATE SEQUENCE employee_seq START WITH 1 INCREMENT BY 1;

CREATE TABLE EMPLOYEE (
   emp_ID          NUMBER DEFAULT employee_seq.NEXTVAL PRIMARY KEY,
   fname           VARCHAR2(50)        NOT NULL,
   lname           VARCHAR2(50)        NOT NULL,
   gender          CHAR(1)             CHECK (gender IN ('M', 'F')),
   age             NUMBER(3),
   emp_email       VARCHAR2(100)       NOT NULL UNIQUE,
   emp_pass        VARCHAR2(255)       NOT NULL,
   job_ID          NUMBER              NOT NULL,
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   salary_ID       NUMBER              NOT NULL,
   CONSTRAINT fk_emp_job    FOREIGN KEY (job_ID)
      REFERENCES JOB_DEPARTMENT(job_ID),
   CONSTRAINT fk_emp_salary FOREIGN KEY (salary_ID)
      REFERENCES SALARY_BONUS(salary_ID)
);


-- TABLE 4: PAYROLL


CREATE SEQUENCE payroll_seq START WITH 1 INCREMENT BY 1;

CREATE TABLE PAYROLL (
   payroll_ID      NUMBER DEFAULT payroll_seq.NEXTVAL PRIMARY KEY,
   pay_date        DATE               NOT NULL,
   report          VARCHAR2(255),
   total_amount    NUMBER(10,2)       NOT NULL,
   emp_ID          NUMBER             NOT NULL,
   salary_ID       NUMBER             NOT NULL,
   CONSTRAINT fk_pay_emp    FOREIGN KEY (emp_ID)
      REFERENCES EMPLOYEE(emp_ID),
   CONSTRAINT fk_pay_salary FOREIGN KEY (salary_ID)
      REFERENCES SALARY_BONUS(salary_ID)
);


-- FK PAYROLL_ID TO JOB_DEPARTMENT

ALTER TABLE JOB_DEPARTMENT
   ADD CONSTRAINT fk_jd_payroll FOREIGN KEY (payroll_ID)
      REFERENCES PAYROLL(payroll_ID);


-- TABLE 5: LEAVE


CREATE SEQUENCE leave_seq START WITH 1 INCREMENT BY 1;

CREATE TABLE LEAVE (
   leave_ID        NUMBER DEFAULT leave_seq.NEXTVAL PRIMARY KEY,
   leave_date      DATE               NOT NULL,
   reason          VARCHAR2(500),
   emp_ID          NUMBER             NOT NULL,
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    payroll_ID      NUMBER,
    CONSTRAINT fk_leave_emp     FOREIGN KEY (emp_ID)
        REFERENCES EMPLOYEE(emp_ID),
    CONSTRAINT fk_leave_payroll FOREIGN KEY (payroll_ID)
        REFERENCES PAYROLL(payroll_ID)
);


-- TABLE 6: QUALIFICATION

CREATE SEQUENCE qualification_seq START WITH 1 INCREMENT BY 1;

CREATE TABLE QUALIFICATION (
    qual_ID         NUMBER DEFAULT qualification_seq.NEXTVAL PRIMARY KEY,
    position        VARCHAR2(100)       NOT NULL,
    date_in         DATE                NOT NULL,
    emp_ID          NUMBER              NOT NULL,
    CONSTRAINT fk_qual_emp FOREIGN KEY (emp_ID)
        REFERENCES EMPLOYEE(emp_ID)
);


-- TABLE 7: EMPLOYEE_ADDRESS

CREATE TABLE EMPLOYEE_ADDRESS (
    emp_ID          NUMBER              NOT NULL,
    contact_add     VARCHAR2(255)       NOT NULL,
    CONSTRAINT pk_emp_address PRIMARY KEY (emp_ID, contact_add),
    CONSTRAINT fk_addr_emp FOREIGN KEY (emp_ID)
        REFERENCES EMPLOYEE(emp_ID)
);


-- TABLE 8: QUALIFICATION_REQUIREMENTS

CREATE TABLE QUALIFICATION_REQUIREMENTS (
    qual_ID         NUMBER              NOT NULL,
    requirements    VARCHAR2(255)       NOT NULL,
    CONSTRAINT pk_qual_req PRIMARY KEY (qual_ID, requirements),
    CONSTRAINT fk_req_qual FOREIGN KEY (qual_ID)
        REFERENCES QUALIFICATION(qual_ID)
);

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02 Normalization

Task 1:

EMP_RAW (emp_id, full_name, phone_numbers, dept_name, dept_location, job_titles, salary)

a) The attributes phone_numbers and job_tittles are multivalued attributes, an employee could have more than one phone number or more than one job title, as well as full_name since it will contain more than one name, and the first normal form states that we should eliminate repeating groups and ensure atomic (indivisible) values in each field.

b) EMP (emp_id, fname,lname, dept_name, dept_location, salary)
EMP (1, Tibyan, Saad, HR, muscat, 650)
EMP (2, Razan, Al-Harthi, HR, muscat, 650)
EMP (3, Mariya, Al-Lamki, IT, sohar, 700)
EMP_PHONE (emp_id,phone_number)
EMP_PHONE (1,98756521)
EMP_PHONE (1,73892423)
EMP_PHONE (2,7365433)
EMP_PHONE (3,98881234)
EMP_JOBS (emp_id , job_title)
EMP_JOBS (1, Recruitment)
EMP_JOBS (2, Records Manager)
EMP_JOBS (1, trainee)
EMP_JOBS (3, Network configuration)

c) My EMS schema does satisfy the 1NF as I have removed all the multivalued attributes and placed them in separate tables that link to the main table to ensure atomic values. Here are the examples:
EMPLOYEE (PK_emp_ID, fname,lname, gender, age, emp_email, emp_pass, FK_job_ID, FK,salary_ID)
EMPLOYEE_ADDRESS (PK_FK_emp_ID, PK_contact_add)
QUALIFICATION_REQUIREMENTS (PK_FK_qual_ID, PK_requirements)
QUALIFICATION (PK_qual_ID, position,date_in,FK_emp_ID)

Task 2:

PAYROLL_RAW (emp_ID, job_ID, emp_name, job_title, salary_range, pay_date, total_amount)

a) Emp_name depends on emp_id
job_title depends on job_id
Salary_range depends on job_id

b) PAYROLL_EMP (emp_ID,emp_name)
PAYROLL_JOB(job_ID, job_title, salary_range)
EMP_JOB(job_ID, emp_ID, pay_date, total_amount)

03-SQL DML — INSERT / UPDATE / DELETE / SELECT

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-- ================================================
-- Task 1
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-- Sequences
CREATE SEQUENCE job_dept_seq START WITH 1 INCREMENT BY 1;
CREATE SEQUENCE salary_bonus_seq START WITH 1 INCREMENT BY 1;
CREATE SEQUENCE employee_seq START WITH 1 INCREMENT BY 1;
CREATE SEQUENCE payroll_seq START WITH 1 INCREMENT BY 1;
CREATE SEQUENCE leave_seq START WITH 1 INCREMENT BY 1;
CREATE SEQUENCE qualification_seq START WITH 1 INCREMENT BY 1;

-- 5 departments
INSERT INTO JOB_DEPARTMENT (job_dept, name, description, salary_range)
VALUES ('HR', 'Human Resources', 'Manages employee relations and recruitment',
'3000-6000');

INSERT INTO JOB_DEPARTMENT (job_dept, name, description, salary_range)
VALUES ('IT', 'Information Technology', 'Manages systems, networks and software
development', '4000-9000');

INSERT INTO JOB_DEPARTMENT (job_dept, name, description, salary_range)
VALUES ('FIN', 'Finance', 'Handles budgeting, accounting and financial reporting', '3500-7000');

INSERT INTO JOB_DEPARTMENT (job_dept, name, description, salary_range)
VALUES ('OPS', 'Operations', 'Oversees daily business operations and logistics', '3000-6500');

INSERT INTO JOB_DEPARTMENT (job_dept, name, description, salary_range)
VALUES ('MKT', 'Marketing', 'Manages branding, campaigns and market research',
'3000-7000');

-- 5 salary records
INSERT INTO SALARY_BONUS (amount, annual, bonus, job_ID)
VALUES (4500, 54000, 1000, 1);

INSERT INTO SALARY_BONUS (amount, annual, bonus, job_ID)
VALUES (7000, 84000, 2000, 2);
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INSERT INTO SALARY_BONUS (amount, annual, bonus, job_ID)
VALUES (5500, 66000, 1500, 3);

INSERT INTO SALARY_BONUS (amount, annual, bonus, job_ID)
VALUES (5000, 60000, 1200, 4);

INSERT INTO SALARY_BONUS (amount, annual, bonus, job_ID)
VALUES (5200, 62400, 1300, 5);

-- 10 employees
INSERT INTO EMPLOYEE (fname, lname, gender, age, emp_email, emp_pass, job_ID,
salary_ID)
VALUES ('Mohammed', 'Al-Rashidi', 'M', 35, 'mohammed.rashidi@ems.com', 'Pass@1234', 1,
1);

INSERT INTO EMPLOYEE (fname, lname, gender, age, emp_email, emp_pass, job_ID,
salary_ID)
VALUES ('Fatima', 'Al-Balushi', 'F', 28, 'fatima.balushi@ems.com', 'Pass@1234', 2, 2);

INSERT INTO EMPLOYEE (fname, lname, gender, age, emp_email, emp_pass, job_ID,
salary_ID)
VALUES ('Ahmed', 'Al-Harthi', 'M', 42, 'ahmed.harthi@ems.com', 'Pass@1234', 3, 3);

INSERT INTO EMPLOYEE (fname, lname, gender, age, emp_email, emp_pass, job_ID,
salary_ID)
VALUES ('Maryam', 'Al-Zadjali', 'F', 31, 'maryam.zadjali@ems.com', 'Pass@1234', 4, 4);

INSERT INTO EMPLOYEE (fname, lname, gender, age, emp_email, emp_pass, job_ID,
salary_ID)
VALUES ('Khalid', 'Al-Farsi', 'M', 38, 'khalid.farsi@ems.com', 'Pass@1234', 5, 5);

INSERT INTO EMPLOYEE (fname, lname, gender, age, emp_email, emp_pass, job_ID,
salary_ID)
VALUES ('Noura', 'Al-Habsi', 'F', 26, 'noura.habsi@ems.com', 'Pass@1234', 1, 1);

INSERT INTO EMPLOYEE (fname, lname, gender, age, emp_email, emp_pass, job_ID,
salary_ID)
VALUES ('Omar', 'Al-Maskari', 'M', 33, 'omar.maskari@ems.com', 'Pass@1234', 2, 2);

INSERT INTO EMPLOYEE (fname, lname, gender, age, emp_email, emp_pass, job_ID,
salary_ID)
VALUES ('Aisha', 'Al-Lawati', 'F', 29, 'aisha.lawati@ems.com', 'Pass@1234', 3, 3);
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INSERT INTO EMPLOYEE (fname, lname, gender, age, emp_email, emp_pass, job_ID,
salary_ID)
VALUES ('Yousuf', 'Al-Mamari', 'M', 45, 'yousuf.mamari@ems.com', 'Pass@1234', 4, 4);

INSERT INTO EMPLOYEE (fname, lname, gender, age, emp_email, emp_pass, job_ID,
salary_ID)
VALUES ('Hessa', 'Al-Siyabi', 'F', 30, 'hessa.siyabi@ems.com', 'Pass@1234', 5, 5);

-- 8 payroll records
INSERT INTO PAYROLL (pay_date, report, total_amount, emp_ID, salary_ID)
VALUES (TO_DATE('2025-01-31', 'YYYY-MM-DD'), 'January payroll processed', 5500, 1, 1);

INSERT INTO PAYROLL (pay_date, report, total_amount, emp_ID, salary_ID)
VALUES (TO_DATE('2025-01-31', 'YYYY-MM-DD'), 'January payroll processed', 9000, 2, 2);

INSERT INTO PAYROLL (pay_date, report, total_amount, emp_ID, salary_ID)
VALUES (TO_DATE('2025-01-31', 'YYYY-MM-DD'), 'January payroll processed', 7000, 3, 3);

INSERT INTO PAYROLL (pay_date, report, total_amount, emp_ID, salary_ID)
VALUES (TO_DATE('2025-01-31', 'YYYY-MM-DD'), 'January payroll processed', 6200, 4, 4);

INSERT INTO PAYROLL (pay_date, report, total_amount, emp_ID, salary_ID)
VALUES (TO_DATE('2025-01-31', 'YYYY-MM-DD'), 'January payroll processed', 6500, 5, 5);

INSERT INTO PAYROLL (pay_date, report, total_amount, emp_ID, salary_ID)
VALUES (TO_DATE('2025-02-28', 'YYYY-MM-DD'), 'February payroll processed', 5500, 1, 1);

INSERT INTO PAYROLL (pay_date, report, total_amount, emp_ID, salary_ID)
VALUES (TO_DATE('2025-02-28', 'YYYY-MM-DD'), 'February payroll processed', 9000, 2, 2);

INSERT INTO PAYROLL (pay_date, report, total_amount, emp_ID, salary_ID)
VALUES (TO_DATE('2025-02-28', 'YYYY-MM-DD'), 'February payroll processed', 7000, 3, 3);

-- 5 leave records
INSERT INTO LEAVE (leave_date, reason, emp_ID, payroll_ID)
VALUES (TO_DATE('2025-01-10', 'YYYY-MM-DD'), 'Medical leave for surgery recovery', 1, 1);

INSERT INTO LEAVE (leave_date, reason, emp_ID, payroll_ID)
VALUES (TO_DATE('2025-01-15', 'YYYY-MM-DD'), 'Annual vacation leave', 2, 2);

INSERT INTO LEAVE (leave_date, reason, emp_ID, payroll_ID)
VALUES (TO_DATE('2025-02-05', 'YYYY-MM-DD'), 'Family emergency', 3, 3);

INSERT INTO LEAVE (leave_date, reason, emp_ID, payroll_ID)
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VALUES (TO_DATE('2025-02-12', 'YYYY-MM-DD'), 'Maternity leave', 4, 4);

INSERT INTO LEAVE (leave_date, reason, emp_ID, payroll_ID)
VALUES (TO_DATE('2025-02-20', 'YYYY-MM-DD'), 'Sick leave', 5, 5);

-- 5 qualification records
INSERT INTO QUALIFICATION (position, date_in, emp_ID)
VALUES ('HR Manager', TO_DATE('2018-03-15', 'YYYY-MM-DD'), 1);

INSERT INTO QUALIFICATION (position, date_in, emp_ID)
VALUES ('Senior Developer', TO_DATE('2019-06-01', 'YYYY-MM-DD'), 2);

INSERT INTO QUALIFICATION (position, date_in, emp_ID)
VALUES ('Financial Analyst', TO_DATE('2017-09-20', 'YYYY-MM-DD'), 3);

INSERT INTO QUALIFICATION (position, date_in, emp_ID)
VALUES ('Operations Supervisor', TO_DATE('2020-01-10', 'YYYY-MM-DD'), 4);

INSERT INTO QUALIFICATION (position, date_in, emp_ID)
VALUES ('Marketing Specialist', TO_DATE('2021-05-25', 'YYYY-MM-DD'), 5);

-- 10 EMPLOYEE_ADDRESS records
INSERT INTO EMPLOYEE_ADDRESS (emp_ID, contact_add)
VALUES (1, 'Muscat, Al Khuwair, Street 12, Villa 5');

INSERT INTO EMPLOYEE_ADDRESS (emp_ID, contact_add)
VALUES (2, 'Muscat, Madinat Al Sultan Qaboos, Street 7, Flat 3');

INSERT INTO EMPLOYEE_ADDRESS (emp_ID, contact_add)
VALUES (3, 'Muscat, Ruwi, Street 4, Building 9');

INSERT INTO EMPLOYEE_ADDRESS (emp_ID, contact_add)
VALUES (4, 'Muscat, Qurum, Street 18, Villa 2');

INSERT INTO EMPLOYEE_ADDRESS (emp_ID, contact_add)
VALUES (5, 'Muscat, Al Azaiba, Street 22, Flat 7');

INSERT INTO EMPLOYEE_ADDRESS (emp_ID, contact_add)
VALUES (6, 'Muscat, Bowsher, Street 3, Villa 11');

INSERT INTO EMPLOYEE_ADDRESS (emp_ID, contact_add)
VALUES (7, 'Muscat, Al Ghubrah, Street 9, Flat 1');

INSERT INTO EMPLOYEE_ADDRESS (emp_ID, contact_add)
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VALUES (8, 'Muscat, Wattayah, Street 6, Building 4');

INSERT INTO EMPLOYEE_ADDRESS (emp_ID, contact_add)
VALUES (9, 'Muscat, Al Mabelah, Street 14, Villa 8');

INSERT INTO EMPLOYEE_ADDRESS (emp_ID, contact_add)
VALUES (10, 'Muscat, Al Hail, Street 5, Flat 2');

-- 5 QUALIFICATION_REQUIREMENTS records
INSERT INTO QUALIFICATION_REQUIREMENTS (qual_ID, requirements)
VALUES (1, 'Bachelor degree in Human Resources or Business Administration');

INSERT INTO QUALIFICATION_REQUIREMENTS (qual_ID, requirements)
VALUES (2, 'Bachelor degree in Computer Science or Software Engineering');

INSERT INTO QUALIFICATION_REQUIREMENTS (qual_ID, requirements)
VALUES (3, 'Bachelor degree in Finance or Accounting');

INSERT INTO QUALIFICATION_REQUIREMENTS (qual_ID, requirements)
VALUES (4, 'Bachelor degree in Business Management or Operations');

INSERT INTO QUALIFICATION_REQUIREMENTS (qual_ID, requirements)
VALUES (5, 'Bachelor degree in Marketing or Communications');

-- Filling JOB_DEPARTMENT payroll_ID column
UPDATE JOB_DEPARTMENT SET payroll_ID = 1 WHERE job_ID = 1;
UPDATE JOB_DEPARTMENT SET payroll_ID = 2 WHERE job_ID = 2;
UPDATE JOB_DEPARTMENT SET payroll_ID = 3 WHERE job_ID = 3;
UPDATE JOB_DEPARTMENT SET payroll_ID = 4 WHERE job_ID = 4;
UPDATE JOB_DEPARTMENT SET payroll_ID = 5 WHERE job_ID = 5;

COMMIT;


-- ================================================
-- Task 2
-- ================================================

-- Employees between age 25 and 40 ordered by last name
SELECT * FROM EMPLOYEE WHERE AGE>25 AND AGE <40 ORDER BY LNAME ASC;

-- Employees with payroll total amount greater than 5000
SELECT E.FNAME,E.LNAME,J.NAME, P.* FROM EMPLOYEE E
JOIN JOB_DEPARTMENT J ON E.JOB_ID = J.JOB_ID
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JOIN PAYROLL P ON P.PAYROLL_ID = J.PAYROLL_ID
WHERE TOTAL_AMOUNT>5000;

-- Employees who took sick leave
SELECT E.FNAME, E.LNAME , L.REASON FROM EMPLOYEE E
JOIN LEAVE L ON E.EMP_ID=L.EMP_ID WHERE LOWER(L.REASON) LIKE
LOWER('%sick%');

-- Departments and their employees
SELECT D.NAME FROM EMPLOYEE E
JOIN JOB_DEPARTMENT D ON D.JOB_ID = E.JOB_ID WHERE(SELECT JOB_ID FROM
EMPLOYEE);
```

04-Aggregation Functions

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-- 04-Aggregation Functions
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-- ================================================
-- Task 1
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-- 1. Total number of employees
SELECT COUNT(EMP_ID) AS TOTAL_EMPLOYEES FROM EMPLOYEE;

-- 2. Minimum, maximum and average salary amount
SELECT MIN(AMOUNT) AS MIN_SALARY FROM SALARY_BONUS;
SELECT MAX(AMOUNT) AS MAX_SALARY FROM SALARY_BONUS;
SELECT AVG(AMOUNT) AS AVG_SALARY FROM SALARY_BONUS;

-- 3. Total bonus amount
SELECT SUM(BONUS) AS TOTAL_BONUS FROM SALARY_BONUS;


-- ================================================
-- Task 2
-- ================================================

-- 1. Departments where average employee age exceeds 30
SELECT D.NAME, AVG(E.AGE) AS AVERAGE_AGE FROM JOB_DEPARTMENT D
JOIN EMPLOYEE E ON E.JOB_ID = D.JOB_ID
GROUP BY D.JOB_ID, D.NAME
HAVING AVG(E.AGE) > 30;

-- 2. Job titles shared by more than 2 employees
SELECT Q.POSITION, COUNT(Q.EMP_ID) AS EMPLOYEE_COUNT FROM QUALIFICATION
Q
GROUP BY Q.POSITION
HAVING COUNT(Q.EMP_ID) > 2;

-- 3. Months where total payroll amount exceeds 20000
SELECT TO_CHAR(PAY_DATE, 'MONTH') AS MONTH_YEAR,
SUM(TOTAL_AMOUNT) AS TOTAL_PAYROLL FROM PAYROLL
GROUP BY TO_CHAR(PAY_DATE, 'MONTH')
HAVING SUM(TOTAL_AMOUNT) > 20000;
```

05-Joins

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-- 05-Joins
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-- ================================================
-- Task 1
-- ================================================

-- Complete employee profile with latest leave date
SELECT E.EMP_ID, E.FNAME || ' ' || E.LNAME AS FULL_NAME, D.NAME AS
DEPARTMENT_NAME,
Q.POSITION AS JOB_TITLE, S.AMOUNT AS SALARY_AMOUNT,
MAX(L.LEAVE_DATE) AS LATEST_LEAVE_DATE FROM EMPLOYEE E
INNER JOIN JOB_DEPARTMENT D ON E.JOB_ID = D.JOB_ID
INNER JOIN SALARY_BONUS S ON E.SALARY_ID = S.SALARY_ID
INNER JOIN QUALIFICATION Q ON E.EMP_ID = Q.EMP_ID
INNER JOIN PAYROLL P ON E.EMP_ID = P.EMP_ID
INNER JOIN LEAVE L ON E.EMP_ID = L.EMP_ID
GROUP BY E.EMP_ID, E.FNAME || ' ' || E.LNAME, D.NAME, Q.POSITION, S.AMOUNT;


-- ================================================
-- Task 2
-- ================================================

-- 1. Employees who have never taken any leave
SELECT E.EMP_ID, E.FNAME, E.LNAME FROM EMPLOYEE E
LEFT JOIN LEAVE L ON L.EMP_ID = E.EMP_ID
WHERE L.LEAVE_ID IS NULL;

-- 2. Departments that have no salary/bonus records
SELECT J.JOB_ID, J.NAME FROM JOB_DEPARTMENT J
LEFT JOIN SALARY_BONUS S ON S.JOB_ID = J.JOB_ID
WHERE S.JOB_ID IS NULL;
```

06-Subqueries

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-- 06-Subqueries
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-- Task 1
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-- 1. Employees earning above company average salary
SELECT E.EMP_ID, E.FNAME || ' ' || E.LNAME AS FULL_NAME,
D.NAME, S.AMOUNT AS SALARY FROM EMPLOYEE E
JOIN SALARY_BONUS S   ON E.SALARY_ID  = S.SALARY_ID
JOIN JOB_DEPARTMENT D ON E.JOB_ID     = D.JOB_ID
WHERE S.AMOUNT > (SELECT AVG(SB.AMOUNT) FROM SALARY_BONUS SB);

-- 2. Department with the highest total payroll amount
SELECT D.NAME, SUM(P.TOTAL_AMOUNT)
FROM JOB_DEPARTMENT D
JOIN PAYROLL P ON P.PAYROLL_ID = D.PAYROLL_ID
GROUP BY D.NAME
HAVING SUM(P.TOTAL_AMOUNT) = (SELECT MAX(TOTAL_AMOUNT) FROM PAYROLL);


-- ================================================
-- Task 2
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-- 1. Employees in departments with bonus greater than 500
SELECT E.EMP_ID, E.FNAME, E.LNAME FROM EMPLOYEE E
WHERE E.JOB_ID IN (SELECT S.JOB_ID FROM SALARY_BONUS S
WHERE S.BONUS > 500);

-- 2. Employees earning more than ALL salaries in Maintenance department
SELECT E.EMP_ID, E.FNAME, E.LNAME, S.AMOUNT FROM EMPLOYEE E
JOIN SALARY_BONUS S ON E.SALARY_ID = S.SALARY_ID
WHERE S.AMOUNT > ALL (SELECT S2.AMOUNT FROM SALARY_BONUS S2
JOIN JOB_DEPARTMENT D ON D.JOB_ID = S2.JOB_ID
WHERE D.NAME = 'Maintenance');

-- 3. Employees earning more than ANY salary in HR department
SELECT E.EMP_ID, E.FNAME, E.LNAME, S.AMOUNT FROM EMPLOYEE E
JOIN SALARY_BONUS S ON E.SALARY_ID = S.SALARY_ID
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WHERE S.AMOUNT > ANY (SELECT S2.AMOUNT FROM SALARY_BONUS S2
JOIN JOB_DEPARTMENT D ON D.JOB_ID = S2.JOB_ID
WHERE D.NAME = 'Human Resources');
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07 Views

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-- 07 Views
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-- Task 1
-- ================================================

-- View creation
CREATE VIEW VW_EMPLOYEE_SUMMARY AS
SELECT E.EMP_ID,E.FNAME || ' ' || E.LNAME AS
FULL_NAME,E.GENDER,E.AGE,D.NAME,Q.POSITION FROM EMPLOYEE E
JOIN JOB_DEPARTMENT D ON E.JOB_ID = D.JOB_ID
JOIN QUALIFICATION Q ON Q.EMP_ID=E.EMP_ID;

-- (a) Query the view to list all female employees over 30
SELECT * FROM VW_EMPLOYEE_SUMMARY WHERE GENDER ='F' AND AGE>30;


-- (b) Attempt to INSERT a row through the view
INSERT INTO VW_EMPLOYEE_SUMMARY (FULL_NAME, GENDER, AGE, NAME,
POSITION)
VALUES ('Tibyan Saad', 'F', 24, 'IT', 'Trainee');

-- Error starting at line : 8 in command -
-- INSERT INTO VW_EMPLOYEE_SUMMARY (FULL_NAME, GENDER, AGE, NAME,
POSITION) VALUES ('Tibyan Saad', 'F', 24, 'IT', 'Trainee')
-- Error at Command Line : 8 Column : 34
-- Error report -
-- SQL Error: ORA-01733: virtual column not allowed here
-- https://docs.oracle.com/error-help/db/ora-01733/01733. 00000 - "virtual column not allowed
here"
-- *Cause: An attempt was made to use an INSERT, UPDATE, or DELETE statement on an
expression in a view.
-- *Action: INSERT, UPDATE, or DELETE data in the base tables, instead of the view.
-- More Details :
-- https://docs.oracle.com/error-help/db/ora-01733/


-- ================================================
-- Task 2
-- ================================================
```

```
-- View creation
CREATE VIEW VW_PAYROLL_DASHBOARD AS
SELECT P.PAYROLL_ID, E.FNAME || ' ' || E.LNAME AS FULL_NAME,
D.NAME,S.AMOUNT,S.BONUS,
L.REASON,P.PAY_DATE,P.TOTAL_AMOUNT FROM EMPLOYEE E
JOIN JOB_DEPARTMENT D ON D.JOB_ID = E.JOB_ID
JOIN PAYROLL P ON P.PAYROLL_ID =D.PAYROLL_ID
JOIN SALARY_BONUS S ON S.JOB_ID = D.JOB_ID
JOIN LEAVE L ON L.PAYROLL_ID = P.PAYROLL_ID;

-- Query the view to find the top 5 payroll records by total_amount.
SELECT * FROM VW_PAYROLL_DASHBOARD ORDER BY TOTAL_AMOUNT DESC FETCH
FIRST 5 ROWS ONLY;
```

08 Stored Procedures & Functions

```
-- ================================================
-- 08 Stored Procedures & Functions
-- ================================================


-- ================================================
-- Task 1
-- ================================================


-- Create procedure
CREATE OR REPLACE PROCEDURE SP_ADD_EMPLOYEE (
   P_FNAME      IN EMPLOYEE.FNAME%TYPE,
   P_LNAME      IN EMPLOYEE.LNAME%TYPE,
   P_GENDER     IN EMPLOYEE.GENDER%TYPE,
   P_AGE        IN EMPLOYEE.AGE%TYPE,
   P_EMP_EMAIL  IN EMPLOYEE.EMP_EMAIL%TYPE,
   P_EMP_PASS   IN EMPLOYEE.EMP_PASS%TYPE,
   P_JOB_ID     IN EMPLOYEE.JOB_ID%TYPE,
   P_SALARY_ID  IN EMPLOYEE.SALARY_ID%TYPE
)
IS
BEGIN
   INSERT INTO EMPLOYEE (FNAME, LNAME, GENDER, AGE, EMP_EMAIL, EMP_PASS,
JOB_ID, SALARY_ID)
   VALUES (P_FNAME, P_LNAME, P_GENDER, P_AGE, P_EMP_EMAIL, P_EMP_PASS,
P_JOB_ID, P_SALARY_ID);

   DBMS_OUTPUT.PUT_LINE('Employee ' || P_FNAME || ' ' || P_LNAME || ' added
successfully.');

   COMMIT;
EXCEPTION
   -- I already have a unique constraint for email during table creation
   WHEN DUP_VAL_ON_INDEX THEN
      RAISE_APPLICATION_ERROR(-20001, 'Error: Email already exists in the system.');
END;
/


-- Testing
```

```
SET SERVEROUTPUT ON;
EXEC SP_ADD_EMPLOYEE('Salim', 'Al-Kindi', 'M',27,
'salim.kindi@ems.com','Pass@5678',1,1);
EXEC SP_ADD_EMPLOYEE('Salim', 'Al-Kindi', 'M',27,
'salim.kindi@ems.com','Pass@5678',1,1);

-- ERROR at line 1:
-- ORA-20001: Error: Email already exists in the system.
-- ORA-06512: at "PROJ.SP_ADD_EMPLOYEE", line 21
-- ORA-06512: at line 1


-- ================================================
-- Task 2
-- ================================================



-- Create procedure
CREATE OR REPLACE FUNCTION FN_NET_SALARY (
   P_EMP_ID IN EMPLOYEE.EMP_ID%TYPE
)
RETURN NUMBER
IS
   V_NET_SALARY NUMBER;
BEGIN
   SELECT S.AMOUNT + S.BONUS INTO V_NET_SALARY
   FROM EMPLOYEE E
   JOIN SALARY_BONUS S ON E.SALARY_ID = S.SALARY_ID
   WHERE E.EMP_ID = P_EMP_ID;

   RETURN V_NET_SALARY;
EXCEPTION
   WHEN NO_DATA_FOUND THEN
      RETURN NULL;
END;
/

-- using procedure in a select statement
SELECT E.EMP_ID,E.FNAME || ' ' || E.LNAME AS FULL_NAME,FN_NET_SALARY(E.EMP_ID)
AS NET_SALARY
FROM EMPLOYEE E ORDER BY NET_SALARY DESC;
```

09 Triggers

```
-- ================================================
-- 09 Triggers
-- ================================================


-- ================================================
-- Task 1
-- ================================================


-- removing the default sequence added at table creation
ALTER TABLE EMPLOYEE MODIFY (EMP_ID NUMBER);

-- creating sequence
CREATE SEQUENCE EMP_SEQ START WITH 11 INCREMENT BY 1;

-- creating trigger
CREATE OR REPLACE TRIGGER TRG_EMP_ID
BEFORE INSERT ON EMPLOYEE
FOR EACH ROW
WHEN (NEW.EMP_ID IS NULL)
BEGIN
   SELECT EMP_SEQ.NEXTVAL INTO :NEW.EMP_ID FROM DUAL;
END;
/

-- tesing the trigger
INSERT INTO EMPLOYEE (FNAME, LNAME, GENDER, AGE, EMP_EMAIL, EMP_PASS,
JOB_ID, SALARY_ID)
VALUES ('Lina', 'Al-Busaidi', 'F', 27, 'lina.busaidi@ems.com', 'Pass@9012', 2, 2);

INSERT INTO EMPLOYEE (FNAME, LNAME, GENDER, AGE, EMP_EMAIL, EMP_PASS,
JOB_ID, SALARY_ID)
VALUES ('Mariam', 'Al-Riyami', 'F', 26, 'mariam.riyami@ems.com', 'Pass@3344', 3, 3);

-- checking
SELECT * FROM EMPLOYEE;
```

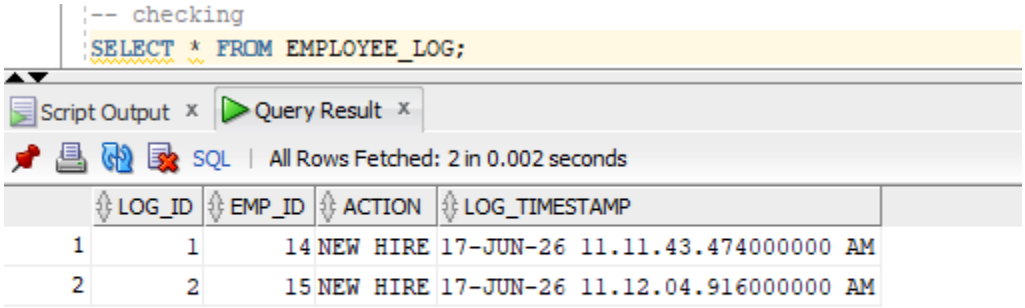
	EMP_ID	FNAME	LNAME	GENDER	AGE	EMP_EMAIL	EMP_PASS	JOB_ID	SALARY_ID
1	1	Mohammed	Al-Rashidi	M	35	mohammed.rashidi@ems.com	Pass@1234	1	1
2	2	Fatima	Al-Balushi	F	28	fatima.balushi@ems.com	Pass@1234	2	2
3	3	Ahmed	Al-Harthi	M	42	ahmed.harthi@ems.com	Pass@1234	3	3
4	4	Maryam	Al-Zadjali	F	31	maryam.zadjali@ems.com	Pass@1234	4	4
5	5	Khalid	Al-Farsi	M	38	khalid.farsi@ems.com	Pass@1234	5	5
6	6	Noura	Al-Habsi	F	26	noura.habsi@ems.com	Pass@1234	1	1
7	7	Omar	Al-Maskari	M	33	omar.maskari@ems.com	Pass@1234	2	2
8	8	Aisha	Al-Lawati	F	29	aisha.lawati@ems.com	Pass@1234	3	3
9	9	Yousuf	Al-Mamari	M	45	yousuf.mamari@ems.com	Pass@1234	4	4
10	10	Hessa	Al-Siyabi	F	30	hessa.siyabi@ems.com	Pass@1234	5	5
11	11	Salim	Al-Kindi	M	27	salim.kindi@ems.com	Pass@5678	1	1
12	13	Mariam	Al-Riyami	F	26	mariam.riyami@ems.com	Pass@3344	3	3
13	12	Lina	Al-Busaidi	F	27	lina.busaidi@ems.com	Pass@9012	2	2

```
-- ================================================
-- Task 2
-- ================================================

-- creating EMPLOYEE_LOG table w/sequence
CREATE SEQUENCE employee_log_seq START WITH 1 INCREMENT BY 1;

CREATE TABLE EMPLOYEE_LOG (
log_id NUMBER DEFAULT employee_log_seq.NEXTVAL PRIMARY KEY,
emp_id NUMBER NOT NULL,
action VARCHAR2(50) NOT NULL,
log_timestamp TIMESTAMP DEFAULT SYSTIMESTAMP,
CONSTRAINT fk_log_emp FOREIGN KEY (emp_id)
REFERENCES EMPLOYEE(emp_id)
);


-- creating trigger
CREATE OR REPLACE TRIGGER TRG_EMP_WELCOME_LOG
AFTER INSERT ON EMPLOYEE
FOR EACH ROW
BEGIN
   INSERT INTO EMPLOYEE_LOG (emp_id, action)
   VALUES (:NEW.EMP_ID, 'NEW HIRE');
END;
/
```

-- inserting records for testing
INSERT INTO EMPLOYEE (FNAME, LNAME, GENDER, AGE, EMP_EMAIL, EMP_PASS, JOB_ID, SALARY_ID)
VALUES ('Yaqoub', 'Al-Amri', 'M', 32, 'yaqoub.amri@ems.com', 'Pass@7788', 4, 4);

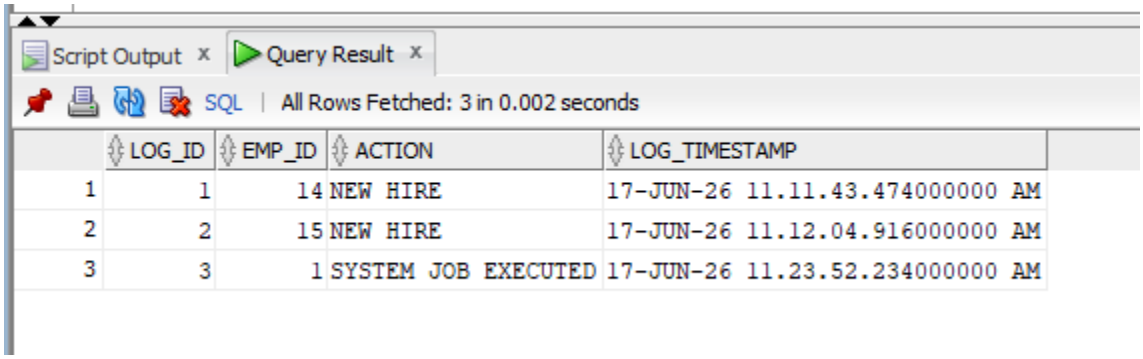
INSERT INTO EMPLOYEE (FNAME, LNAME, GENDER, AGE, EMP_EMAIL, EMP_PASS, JOB_ID, SALARY_ID)
VALUES ('Salma', 'Al-Wahaibi', 'F', 29, 'salma.wahaibi@ems.com', 'Pass@4455', 5, 5);

-- checking
SELECT * FROM EMPLOYEE_LOG;



10 Oracle Scheduler Jobs

```
-- ================================================
-- 10 Oracle Scheduler Jobs
-- ================================================



-- ================================================
-- Task 1
-- ================================================

-- creating the job
BEGIN
   DBMS_SCHEDULER.CREATE_JOB(
      job_name       => 'JOB_GREET_EMPLOYEES',
      job_type       => 'PLSQL_BLOCK',
      job_action     => 'BEGIN
                          DBMS_OUTPUT.PUT_LINE("Payroll System Initialized");
                          INSERT INTO EMPLOYEE_LOG (emp_id, action)
                          VALUES (1, "SYSTEM JOB EXECUTED");
                        END;',
      start_date     => SYSTIMESTAMP + INTERVAL '2' MINUTE, --once in 2 mins
      enabled        => TRUE
   );
END;
/

-- testing after 2 mins
SELECT * FROM EMPLOYEE_LOG;
```



	LOG_ID	EMP_ID	ACTION	LOG_TIMESTAMP
1	1	14	NEW HIRE	17-JUN-26 11.11.43.474000000 AM
2	2	15	NEW HIRE	17-JUN-26 11.12.04.916000000 AM
3	3	1	SYSTEM JOB EXECUTED	17-JUN-26 11.23.52.234000000 AM

```
-- ================================================
-- Task 2
-- ================================================

-- creating job

BEGIN
   DBMS_SCHEDULER.CREATE_JOB(
      job_name        => 'JOB_DAILY_LEAVE_REPORT',
      job_type        => 'PLSQL_BLOCK',
      job_action      => 'DECLARE
                            V_LEAVE_COUNT NUMBER;
                          BEGIN
                            SELECT COUNT(*) INTO V_LEAVE_COUNT
                            FROM LEAVE
                            WHERE TRUNC(LEAVE_DATE) = TRUNC(SYSDATE);
                            INSERT INTO EMPLOYEE_LOG (emp_id, action)
                            VALUES (1, "DAILY LEAVE REPORT: " || V_LEAVE_COUNT || " leave
record(s) today");
                          END;',
      start_date      => SYSTIMESTAMP,
      repeat_interval => 'FREQ=DAILY; BYHOUR=7; BYMINUTE=0; BYSECOND=0',
      enabled         => TRUE
   );
END;
/

-- showing job definition
SELECT * FROM USER_SCHEDULER_JOBS WHERE JOB_NAME =
'JOB_DAILY_LEAVE_REPORT';
```